

IntelliCane: Smart Assistive White Cane

Obstacle Avoidance, Fall Detection, and Companion Mobile App for Caretakers

Team Neural-Nexus | Track A: Embedded IoT System | IoTrix 2.0 Semi-Final (5 Oct 2026)

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Project Repository: <https://github.com/Neural-Nexus-IoTrix-2-0/intellicane>

1. The Problem: Why We Built IntelliCane

A traditional white cane is great for finding steps and curbs on the ground, but it cannot warn visually impaired people about waist- or head-level hazards—like open car doors, tree branches, or table corners. In addition, if a user trips and falls, they can easily get hurt and isolated without any way to alert family. Existing smart canes cost upwards of \$500 (LKR 150,000+), depend on a reliable internet connection, or blast annoying beeps that make it hard to hear surrounding traffic. As undergraduate students, we wanted to build something genuinely affordable, practical, and dependable that works offline while keeping caretakers in the loop.

2. How IntelliCane Works

IntelliCane is built in two simple, connected parts:

- Smart Cane Hardware (ESP32-C3):** The cane works completely on its own without needing a phone or internet. An ultrasonic sensor scans ahead up to 4 m. A small motor in the handle vibrates gently when an obstacle is 60 cm away, and vibrates progressively stronger as you get closer (down to 10 cm). A motion sensor detects sudden falls or drops. During a fall, the cane stays quiet locally (no loud buzzing or alarm to panic the user) and instantly transmits an emergency alert over Bluetooth.
- Companion Mobile App (application/):** Connects to the cane over Bluetooth Low Energy with two modes:
 - Blind User Mode:* Clean high-contrast screen showing connection status, a 1-tap location sharing button, and a collapsible diagnostic terminal for testing.
 - Caretaker Mode:* Shows the user’s live status, displays their real-time location on an interactive map with a quick re-center button, and **sends instant emergency push notifications (with sound and vibration)** to the caretaker’s phone the moment a fall occurs!

3. System Overview

On the Cane Ultrasonic Sensor (Obstacles) Motion Sensor (Tilt & Falls)	→	Microcontroller (ESP32-C3) Fast offline loop (under 50 ms) Fall detection logic	→	Alerts for the User Vibration Motor (Stronger as you get closer) Buzzer (Warning tone; silent during falls)
Wireless Bluetooth Low Energy (BLE) Link				
Companion Mobile App Blind User Dashboard Diagnostic Terminal	→	Cloud Services (Firebase) Live phone GPS location Automatic cloud sync	→	Caretaker Phone Real-time tracking map (1-tap centering) Instant fall alert notification (Sound + Vibrate)

4. What We Have Built & Verified

- Working Hardware Prototype:** Assembled and tested with the ESP32-C3 controller, ultrasonic sensor (Echo wired directly to GPIO 1), motion sensor, handle vibration motor, and buzzer for just LKR 2,800.
- Intuitive Handle Vibration:** The handle vibrates smoothly between 60 cm and 10 cm. It stays off past 60 cm so the user’s hand does not get tired.
- Silent Fall Protection:** If the cane tilts past 65° or takes a hard drop, it keeps quiet on the cane and immediately alerts the phone over Bluetooth. Standing the cane back up (< 30°) automatically clears the emergency and resumes normal guidance.
- Caretaker Alerts & Automated SMS:** Built in Kotlin (application/) with a live tracking map, vibrating emergency push alerts, and automated SMS emergency alerts to emergency contacts the moment a fall is detected.
- Bench Testing:** Verified ultrasonic distance accuracy with a physical tape measure from 2 cm to 2.5 m (±1.5 cm accuracy). Tested fall detection with a digital protractor and ran software unit tests to ensure calculations never crash or freeze.

5. Components & Cost (Total: LKR 2,800)

- Cane Electronics:** ESP32-C3 SuperMini (LKR 1,400), Ultrasonic Sensor (LKR 450), Motion Sensor (LKR 600), Vibration Motor (LKR 250), Buzzer & Jumpers (LKR 100). **Total Cost: LKR 2,800.**
- Software Tools:** C++ firmware (Arduino-ESP32), Android app in Kotlin (application/), Firebase cloud database, Open-StreetMap / ESRI live maps.

6. Known Limitations & Roadmap to the Final (17 October 2026)

- Current Limitations & Solutions:** Soft angled clothing can sometimes scatter ultrasound beyond 2 m (we will add a compact laser sensor for the final). Breadboard jumper wires can loosen during heavy movement (we are making a custom soldered PCB and 3D-printed handle).
- Next Steps for the Final:** (1) Build the custom soldered PCB and ergonomic handle with TPU shock damping, (2) Add a rechargeable 18650 battery with charge protection for all-day runtime, and (3) Add a secondary Time-of-Flight laser sensor.